UNIVERSITI TENAGA NASIONAL

The Energy University

CGEB4112 · ENERGY & COMPUTING · SECTION 1

Blockchain in the Oil & Gas Industry

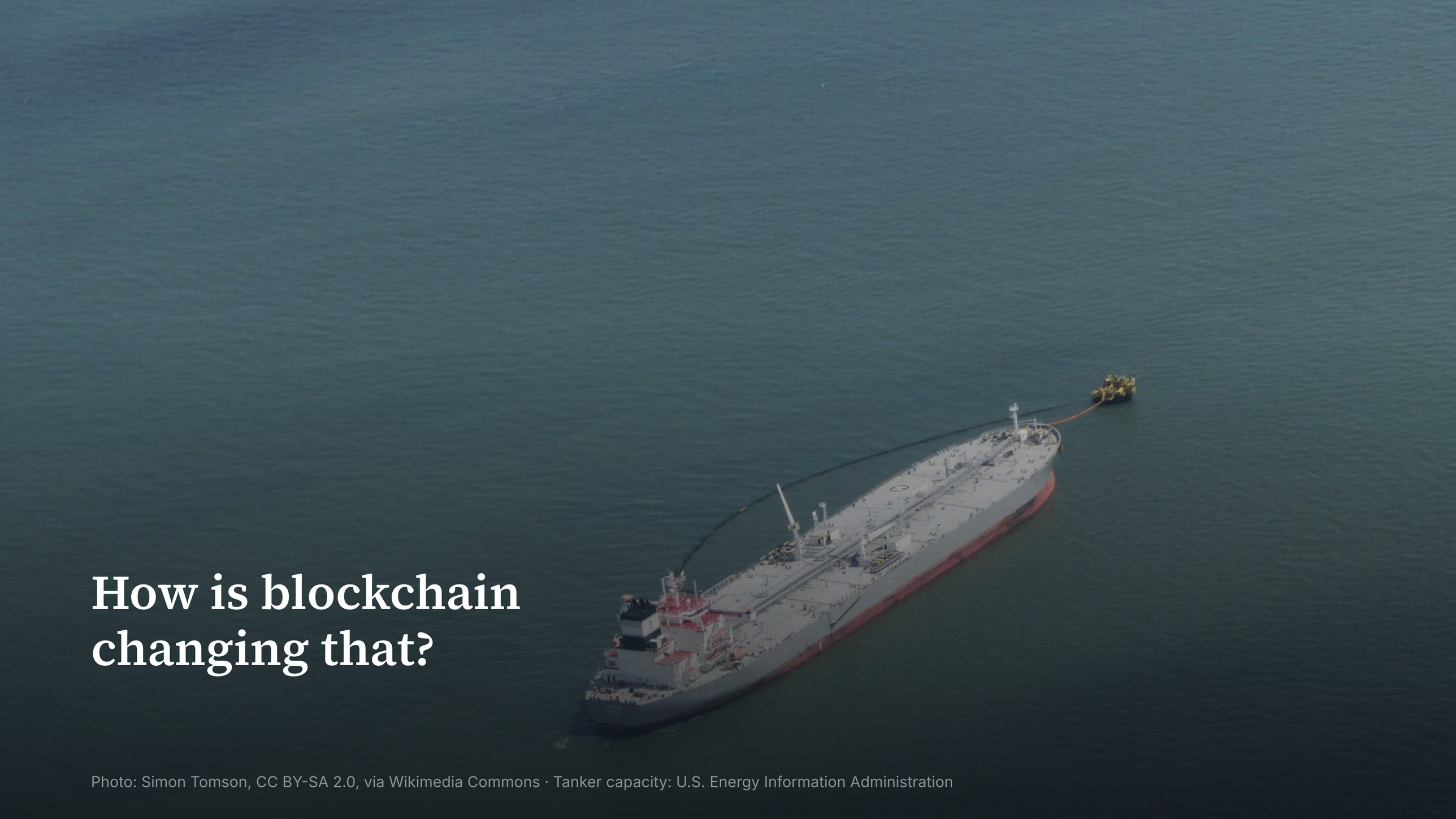
GROUP 10

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How is blockchain changing that?

Photo: Simon Tomson, CC BY-SA 2.0, via Wikimedia Commons · Tanker capacity: U.S. Energy Information Administration

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Blockchain in the Oil & Gas Industry

01 How the oil business works

02 What blockchain is

03 Where it’s used

04 Real example: VAKT

05 Challenges & benefits

Three stages of oil

Petronas: all three



Upstream

Find · Drill · Produce



Midstream

Transport · Store

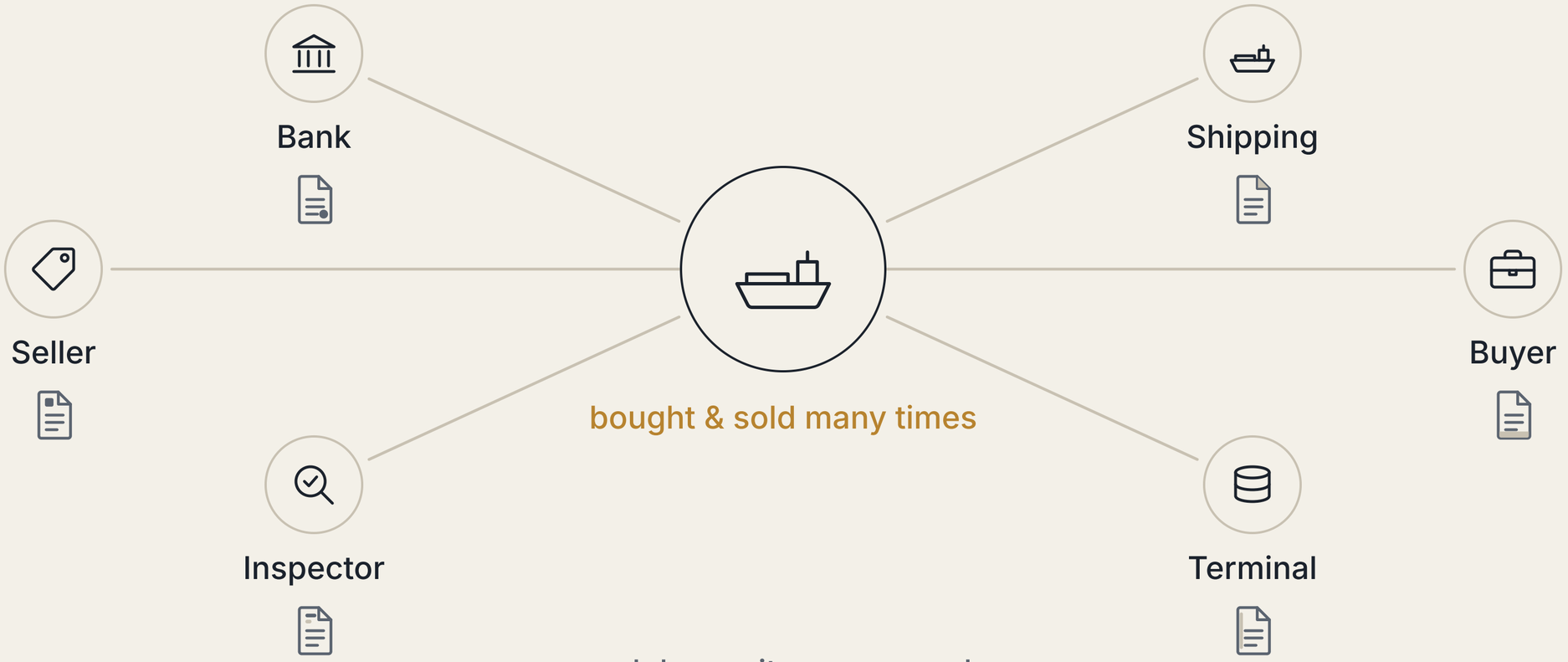


Downstream

Refine · Sell

Photos via Wikimedia Commons: Lindsey G (CC BY 2.0) · Arthur T. LaBar (CC BY 2.0) · Carol M. Highsmith, Library of Congress (public domain)

One cargo, many hands



each keeps its own records

No single version of the truth.

Blockchain = one shared record

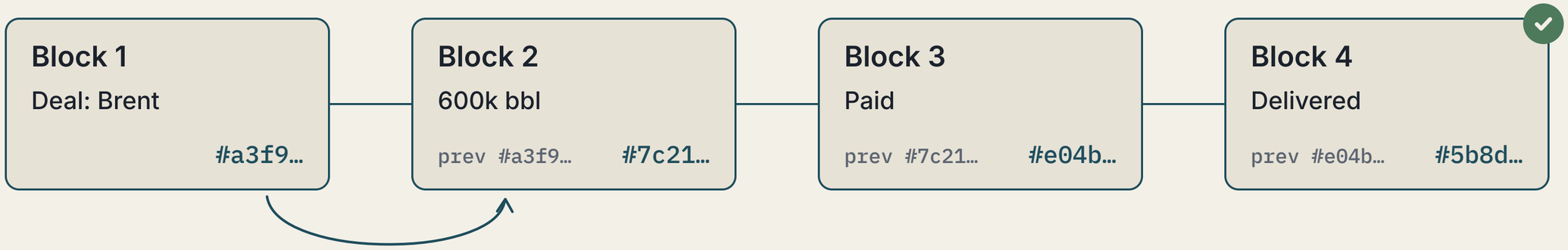
Seller

Buyer

Bank

Shipping

Inspector



add ✓ · secretly change ✗

Source: S. Nakamoto, “Bitcoin: A Peer-to-Peer Electronic Cash System” (2008)

Smart contract

IF	cargo arrives	✓
AND	passes inspection	✓
→ THEN	release payment	



Sources: S. Nakamoto (2008) · N. Szabo, “Formalizing and Securing Relationships on Public Networks” (1997)

Where it’s used



Upstream



Midstream



Downstream

Paying suppliers automatically

Pilot: 10 oil companies incl. Chevron, ExxonMobil, Shell

Platform: GumboNet

water-hauling trucks tracked → bills settled by smart contract

Every handover recorded once

Deals, documents, finance — digital

Origin & carbon footprint tracking

Sources: Data Gumbo (GumboNet) · Blockchain for Energy (formerly OOC Oil & Gas Blockchain Consortium)

When does blockchain make sense?

Not for everything.

- ✓ Many companies share the same data
- ✓ They don’t fully trust each other
- ✓ Mistakes or fraud are expensive

Oil trading ✓✓✓



First market: North Sea crude (Brent)

Sources: VAKT (2017–2020) · Photo: Banja-Frans Mulder, CC BY 3.0, via Wikimedia Commons

VAKT

London · real-world example

Deal agreed → POST-TRADE: confirm · documents · finance · settle

2017–18	2019	2020
BP	Chevron	Saudi Aramco
Shell	Total	$5 M
Equinor	Reliance	

traders & banks

How a deal works

1 Enter trade

2 Match

3 Shipping & inspection

4 Finance

Built on Quorum

private Ethereum, first built by JPMorgan

Trade entered
#a3f9...

Matched
#7c21...

Inspection report
#e04b...

Financed
#5b8d...

only the parties in the deal can see it

Sources: VAKT · komgo · J.P. Morgan (Quorum)

The result





of North Sea oil deals on VAKT at launch (2019)

Still in use — 2024: Shell Trading Rotterdam adopts vSure

Sources: VAKT (launch, 2019; vSure at Shell Trading Rotterdam, 2024)

Challenges

Why it’s not everywhere yet

01 **Adoption**
works only if everyone joins



02 **Integration**
old trading & accounting systems

03 **Law & standards**
e-documents not accepted everywhere

04 **Privacy**
rivals must not see prices

05 **Garbage in, garbage out**
can’t verify data at entry



Benefits

One shared truth

no re-checking

Speed

days ⟶ minutes

Less fraud

no double financing

Audit trail

who · what · when

Automation

smart contracts pay & approve

Thank you

Questions?

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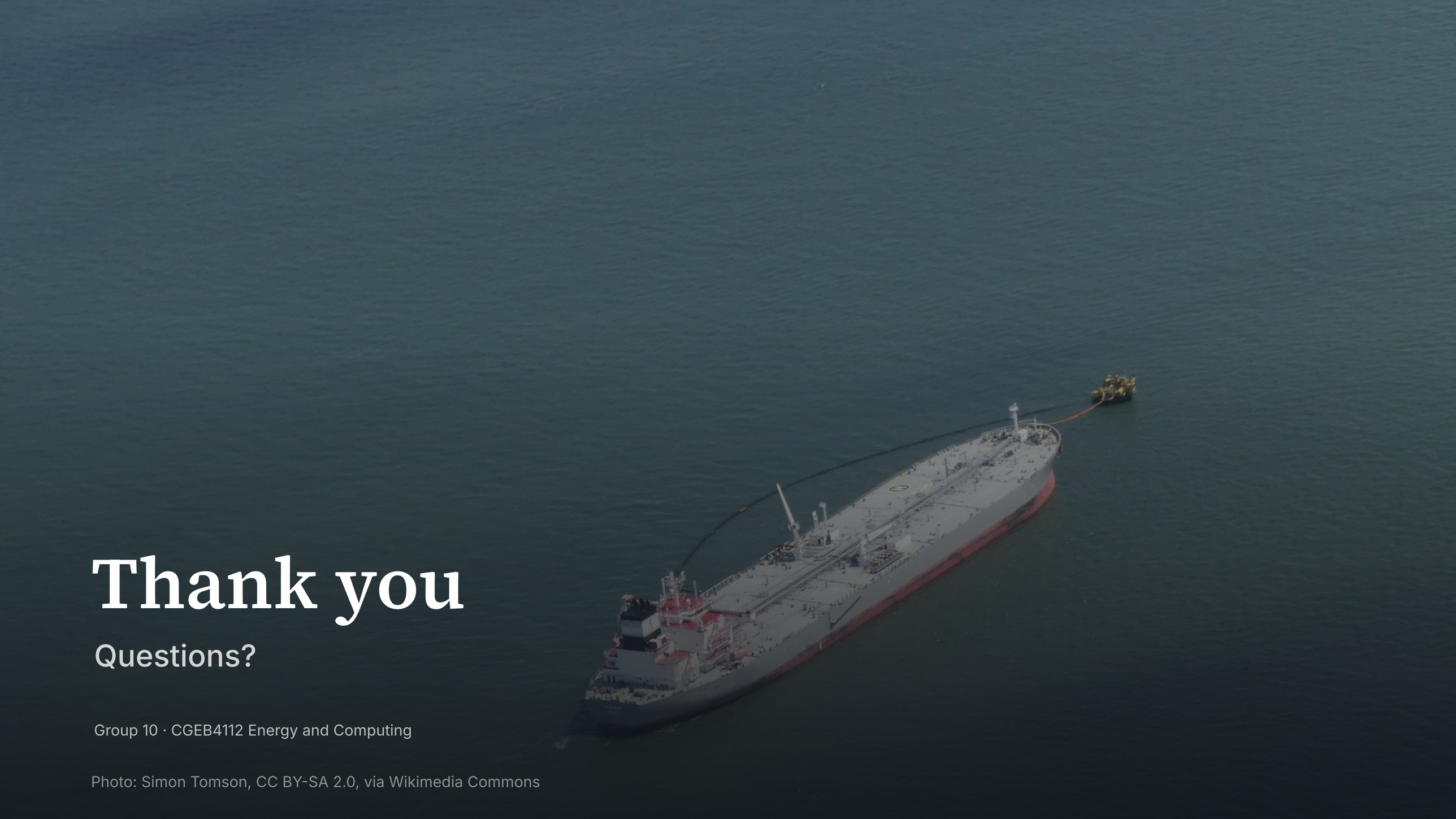


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